Robust Semantic Segmentation under Adverse Weather

Independent Research in Visual Perception under Distribution Shift

Studied how semantic segmentation degrades under severe weather shifts and developed adaptation strategies using feature alignment and vision foundation model priors to improve robustness on the ACDC dataset.

Problem

Semantic segmentation models trained on clear weather urban scenes degrade severely under fog, rain, snow, and night. The degradation pattern differs across conditions, which motivates condition-aware adaptation rather than a single uniform constraint.

Setup

Dataset: ACDC

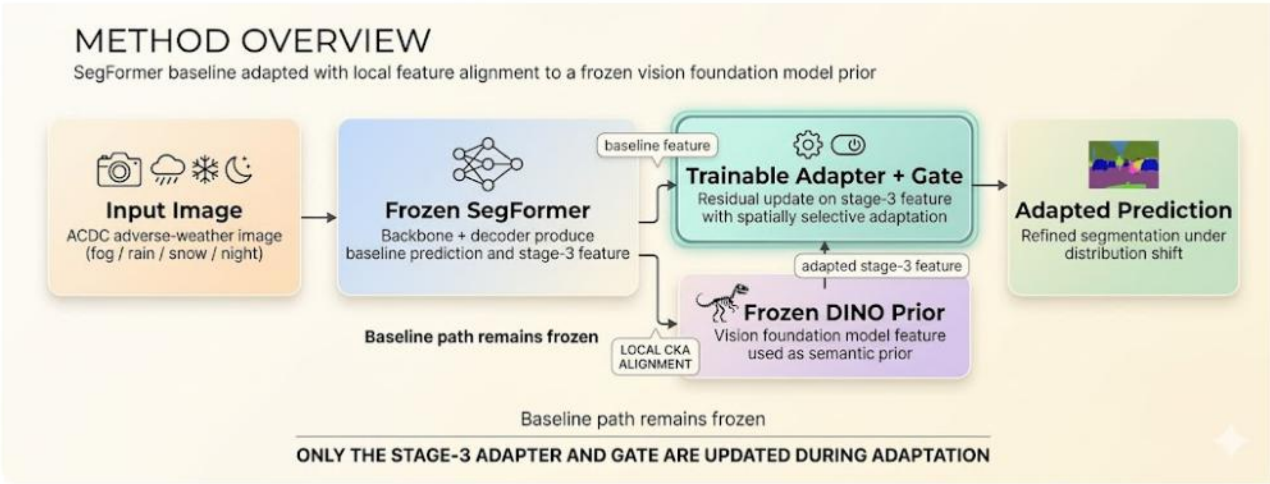
Baseline: SegFormer

Goal: improve robustness under adverse weather distribution shift

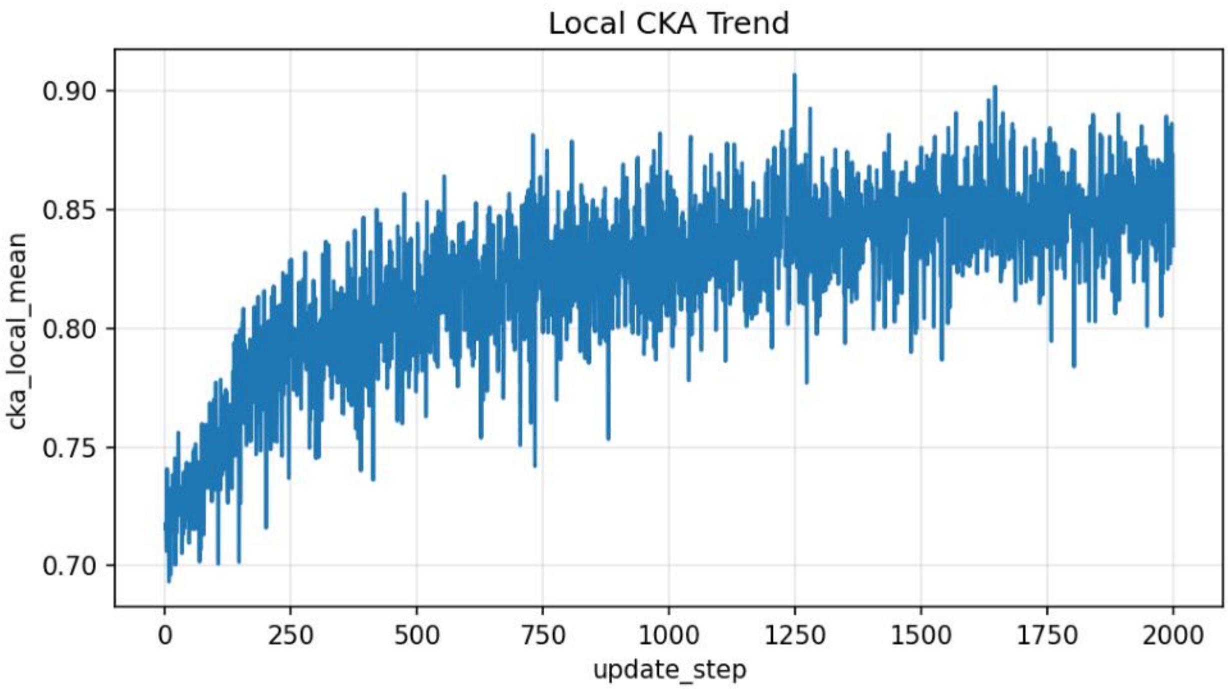
Direction: feature alignment and semantic priors from DINOv3

Method

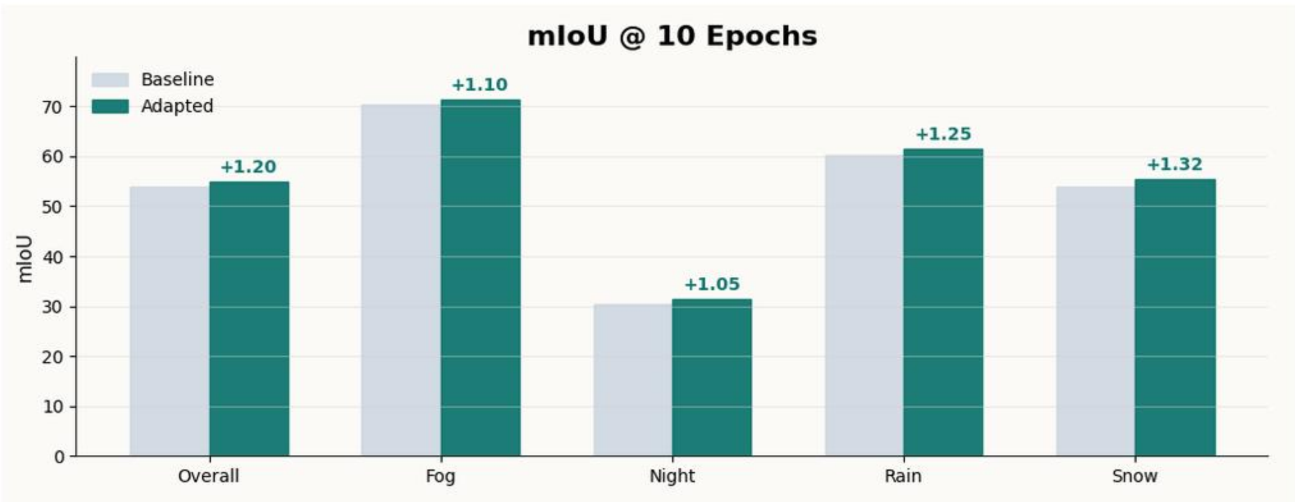
Built a semantic segmentation evaluation pipeline on ACDC with a SegFormer baseline and analyzed robustness under fog, rain, snow, and night. To examine how feature correspondence changes under distribution shift, I applied local CKA over 10 spatial windows and tracked alignment patterns across stages and conditions. Using these diagnostics, I explored adaptation strategies based on feature level alignment and DINOv3 semantic priors to improve robustness while maintaining semantic consistency.



Overall pipline



Local CKA during 5epochs



mIoU after 10 epochs

qualitative results: adapted = yellow, false negative = green

My role

Defined the research question, implemented the training and evaluation pipeline, designed ablations, analyzed failure cases, and reformulated the next research direction.

Next step

- Extending the project toward TTA with lightweight adapters and stronger semantic priors.
- Considering feature channel axis alignment between DINOv3 and SegFormer decoder.